

AI AGENTS IN PRODUCTION

WHAT ACTUALLY MAKES AN AGENT RELIABLE?

A field guide to agent loops, tools, state, policy, recovery, observability and evaluation.

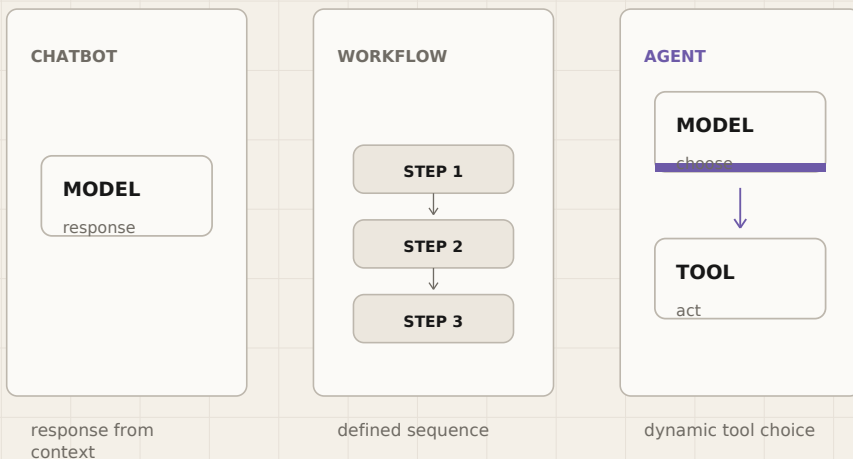
DECISION TRACE

- 01 **TASK**
- 02 **PLAN**
- 03 **ACT**
- 04 **OBSERVE**

CORE RULE

**THE MODEL PROPOSES.
THE SYSTEM ENFORCES.**

CHATBOT · WORKFLOW · AGENT

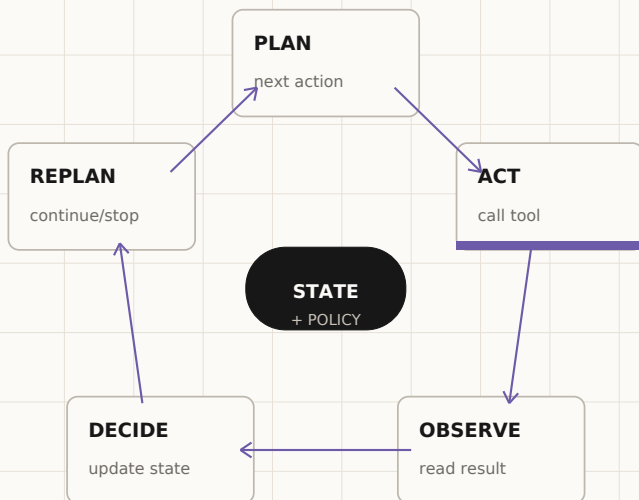


PRODUCTION QUESTION

Who decides the next step?

If it changes at runtime, you need agent controls.

PLAN · ACT · OBSERVE · DECIDE



BOUNDARIES

TIMEOUT

every action has a limit

RETRY

retries are finite

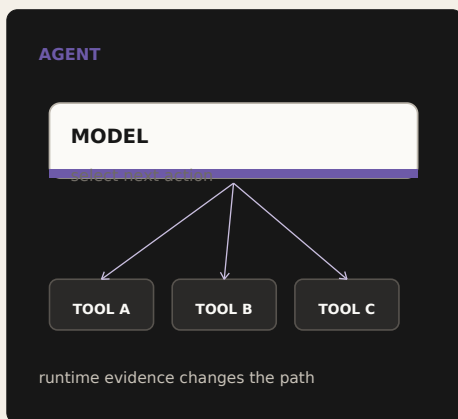
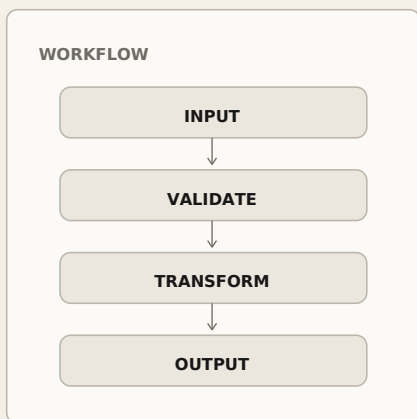
STOP

agent can terminate

ESCALATE

uncertainty can leave

WHEN SHOULD THE SYSTEM BE DYNAMIC?



RULE OF THUMB

Known path → workflow

Unknown next action → agent

THE AGENT SHOULD NEVER CALL A TOOL BLIND



TOOL CONTRACT

name	refund_status
input	order_id: string
permission	read:orders
failure	timeout / not_found
side_effect	none

MAKE THE SAFE PATH THE EASY PATH

AMBIGUOUS

update_customer()

id: ? fields: ? scope: ?

large action surface

The model must infer intent, fields and side effects.

EXPLICIT

change_email()

user_id: string
new_email: string
permission: profile.write

narrow action surface

NARROW

one intent

TYPED

explicit inputs

SCOPED

least privilege

OBSERVABLE

clear result

STATE RULE

Task state is ephemeral. Memory is deliberate. Treat retention, access and deletion as policy.

TASK STATE IS NOT MEMORY

RUNTIME STATE

CURRENT TASK

goal · step · tool result

SESSION CONTEXT

conversation · approvals

RETAINED MEMORY

only durable facts

MEMORY POLICY

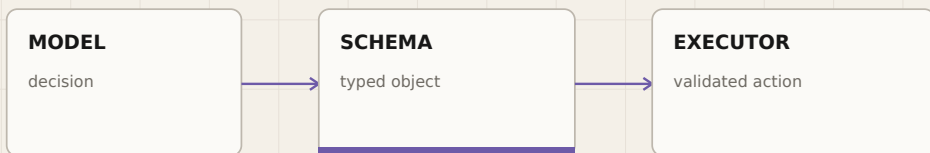
STORE

purpose + retention + access

OUTPUT RULE

A typed object is a contract. Invalid shape, missing field or disallowed action should stop execution.

LET THE SCHEMA DECIDE



EXAMPLE / SHAPE

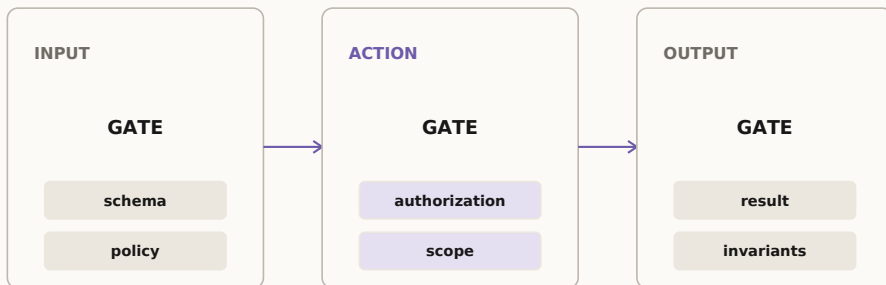
```
{ "action": "refund_status",
  "order_id": "A19",
  "confidence": 0.94,
  "requires_approval": false }
```

invalid shape → reject

GUARDRAIL RULE

Validation belongs at the boundary where damage can still be prevented, not after the tool has already acted.

VALIDATE AT THE BOUNDARY



FAIL CLOSED

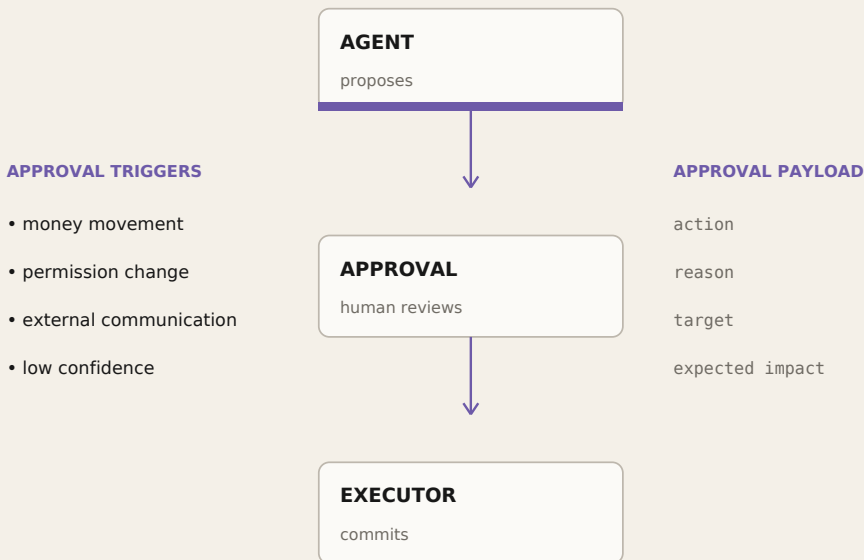
If validation fails, do not silently continue.

Return a structured failure, escalate, or route to a safe fallback.

HUMAN RULE

Approval should carry the action, target, reason and expected impact so the human can make a bounded decision.

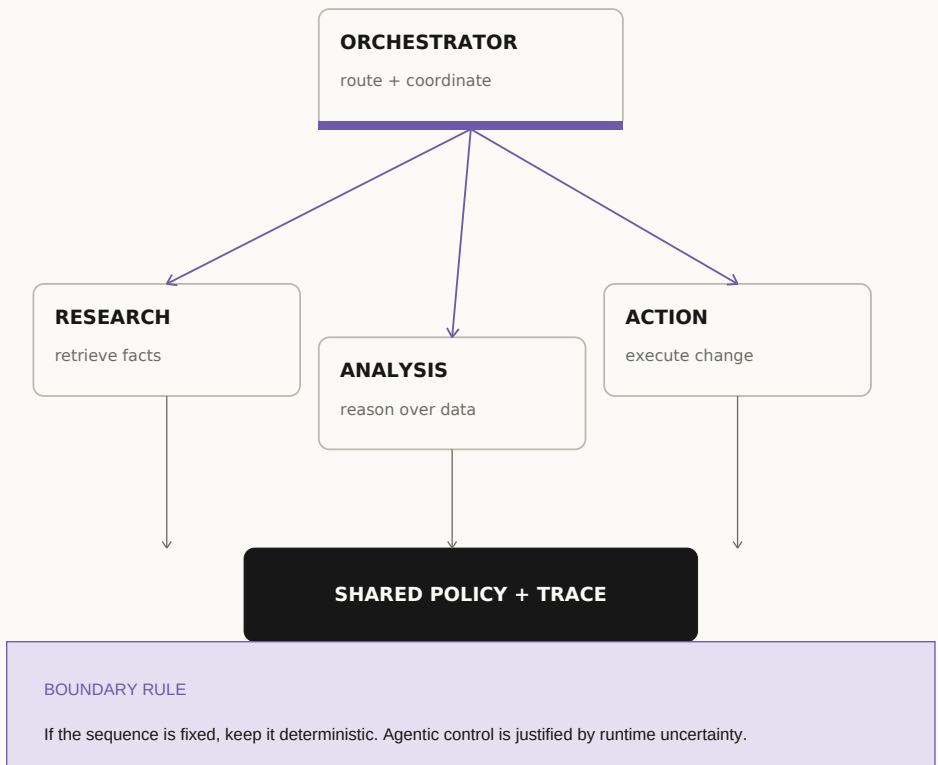
NOT EVERY ACTION SHOULD BE AUTONOMOUS



ORCHESTRATION RULE

Specialists should have narrow responsibilities. Shared policy and tracing remain outside the agents.

MORE AGENTS · MORE INTELLIGENCE



SOMETIMES THE RIGHT AGENT IS NO AGENT

SIGNAL	PREFER	WHY
fixed sequence	WORKFLOW	predictable
strict validation	FUNCTION	deterministic
dynamic tool choice	AGENT	runtime uncertainty
high-impact action	AGENT + HUMAN	control

DESIGN TEST

Can you describe the next step before runtime evidence exists?

Yes → workflow. No → consider an agent.

WHAT BREAKS FIRST

01

WRONG TOOL

valid tool, wrong intent

02

BAD ARGS

schema mismatch

03

STALE STATE

old context drives action

04

FALSE COMPLETE

model claims success

05

RETRY LOOP

same failure repeats

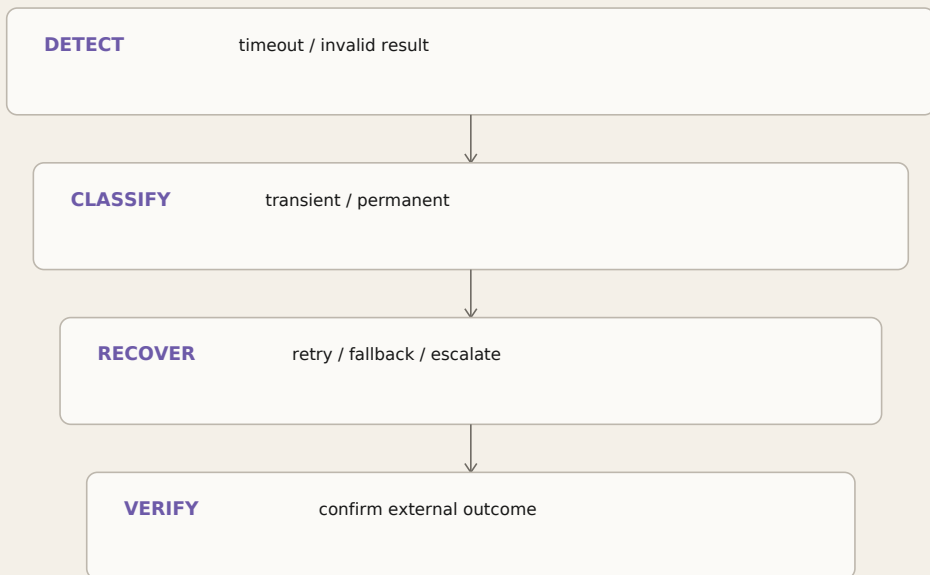
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PARTIAL EXEC

external system fails midway

Every failure needs: **detection → bounded response → observable outcome**

FAILURE SHOULD CHANGE THE NEXT ACTION



SECURITY RULE

Every tool call is an authorization decision, even when the model is the component proposing it.

AN AGENT IS AN IDENTITY WITH ACCESS

IDENTITY

who is acting?

SCOPE

what can it access?

APPROVAL

what needs consent?

AUDIT

what happened?

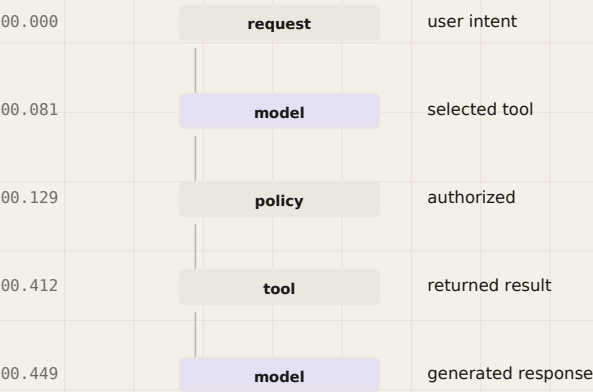
LEAST PRIVILEGE

OBSERVABILITY RULE

Record request, model decision, policy result, tool call, latency and final outcome as one trace.

TRACE THE DECISION, NOT JUST THE RESPONSE

REQUEST TRACE



WHAT TO MEASURE

TOOL	LATENCY	OUTCOME	COST
selection + args	model + tool	task success	tokens + calls

EVALUATION RULE

Score the trajectory: task success, tool choice, arguments, policy compliance, recovery and cost.

EVALUATE THE TRAJECTORY, NOT JUST THE ANSWER

TASK COMPLETION



TOOL SELECTION



ARGUMENT QUALITY



POLICY COMPLIANCE



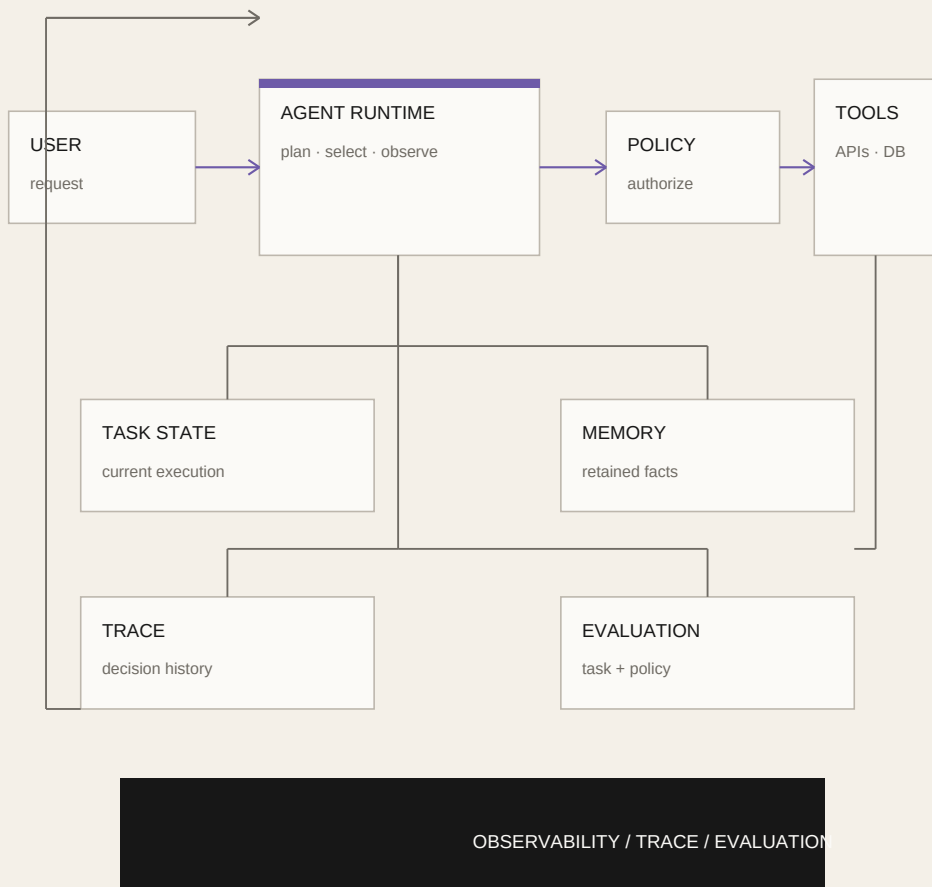
RECOVERY



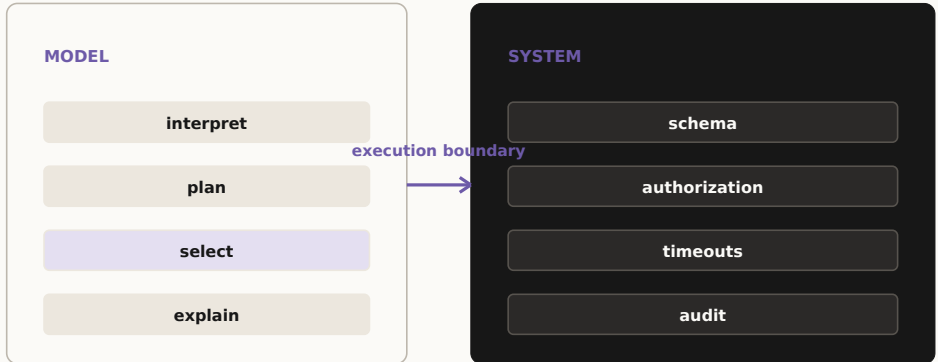
TEST CASES SHOULD INCLUDE:

normal path · ambiguous input · tool failure · policy boundary

A PRODUCTION AGENT IS A SYSTEM



THE MODEL PROPOSES. THE SYSTEM ENFORCES.



DO NOT ASK THE MODEL
to be the policy engine.
Let it reason inside one.

BEFORE YOU SHIP AN AGENT

- ✓

GUARDRAILS

input/action/output validation
- ✓

BUDGETS

latency + token limits
- ✓

AUTHORIZATION

side effects policy checked
- ✓

EVALUATION

trajectories tested
- ✓

SCOPED STATE

task and memory separated
- ✓

TRACING

decisions and outcomes visible
- ✓

TYPED TOOLS

inputs + permissions explicit
- ✓

HUMAN ESCALATION

high-impact actions can stop
- ✓

TASK BOUNDARY

clear job + stop condition
- ✓

RECOVERY

bounded retries + escalation

SHIP GATE

Every item above must map to a real control.

CODE

POLICY

INFRASTRUCTURE

START SMALL. MAKE FAILURE VISIBLE. BOUND EVERYTHING.